

Nadun Ranasinghe

Doctoral Researcher, Tampere University

1. PERSONAL DETAILS AND THE DATE OF THE CV

Surname: Ranasinghe

First names: Nadun

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2. DEGREES

Doctor of Science(Technology) in Robotics and Automation Engineering Reading

Automation Technology and Mechanical Engineering, Tampere University Tampere, Finland

- My work focuses on developing intelligent, interpretable and human-centric AI systems that enable safe, effective, and intuitive collaboration between humans and robots on the factory floor.

- Funded by: Doctoral School of AI (AI-DOC) run by the Finnish Center for Artificial Intelligence (FCAI)

Master of Science (Technology) in Automation Engineering 2024

Automation Technology and Mechanical Engineering, Tampere University Tampere, Finland

- Specialisation: Factory Automation and Robotics.
- Master's thesis: *Multi-agent verbal communication enabling the execution of multiple actions through a single interaction for the next generation of human-robot collaboration.*
- Graduated with Distinction, earning the graduation award from The Industrial Research Fund at Tampere University of Technology

Bachelor of Engineering (Honours) in Electrical and Electronics Engineering 2022

Curtin University Perth, Australia

- Bachelor's Thesis: *Contextual Generative Adversarial Networks based face sketch-to-image synthesis*
- Graduated with a Second Upper Class

3. OTHER EDUCATION AND EXPERTISE

Research expertise: Human-robot collaboration, conversational cobots, multimodal interaction, large language models, cognitive validation, industrial automation, collaborative robotics, multi-robot systems, formation control, and robot programming.

Technical expertise: Python, ROS, MATLAB, ABB robot programming, KUKA robot programming, Omron robot programming, SCADA systems, circuit design, PCB design, cloud-based robotic decision-support systems, and web development.

Document preparation and research communication: LaTeX, Microsoft Office Suite, technical documentation, academic writing, and research web communication.

4. LANGUAGE SKILLS

Sinhala: Native language.

English: C1, professional working proficiency.

Finnish: A1, basic proficiency.

5. CURRENT EMPLOYMENT

Doctoral Researcher

August 2024 – Present

FAST-Laboratory, Faculty of Engineering and Natural Sciences, Tampere University

Tampere, Finland

- Doctoral research in human–robot collaboration, multimodal interaction, conversational cobots, and AI-enabled industrial automation.
- Development of multimodal interaction systems for collaborative robotics in industrial environments.
- Research on large language models in human–robot collaboration, including cognitive validation and mitigation of context-induced hallucinations.
- Stage of academic research career: early-stage researcher / doctoral researcher.

6. PREVIOUS WORK EXPERIENCE

Research Assistant / Researcher

October 2022 – July 2024

FAST-Laboratory, Tampere University

Tampere, Finland

- Designed and implemented the FAST-Laboratory research group website, including project-specific pages.
- Worked on redevelopment tasks for the FASTory research line, involving eight single-arm robot manipulators from manufacturers including SONY, KUKA, Omron, and ABB.
- Contributed to the Open Alliance Pilot, a multimodal interaction research pod for the AI-PRISM project funded by the European Union.
- Completed Master’s thesis research on multi-agent verbal communication for next-generation human–robot collaboration.

Academic Instructor

February 2022 – July 2022

Sri Lanka Institute of Information Technology

Malabe, Sri Lanka

- Conducted tutorials and lectures for courses in the Department of Electrical and Electronic Engineering.
- Supported teaching, student learning, and practical engineering education.
- Participated in organising all-island robotics competitions and workshops.

Research Assistant

January 2021 – January 2022

CSAAT, Sri Lanka Institute of Information Technology

Malabe, Sri Lanka

- Worked on research introducing the NAO V6 robot for autism diagnosis in early childhood in the Sri Lankan context under the CSAAT project, funded by the AHEAD grant of the World Bank.
- Contributed to research involving cloud computing and Partially Observable Markov Decision Processes for developing a robotic decision-support system for diagnostic interaction.

Engineering Intern

November 2018 – February 2019

OREL Corporation

Panaluwa, Sri Lanka

- Worked as a maintenance engineering intern in the cable manufacturing department.
- Gained practical experience in factory control, automation, SCADA, circuit design, and circuit development.
- Developed a SCADA system for boiler heat monitoring using NodeMCU communication.
- Designed, developed, and etched printed circuit boards for an opto-coupler configuration.

7. CAREER BREAKS

No career breaks declared.

8. RESEARCH FUNDING AND GRANTS

AI-DOC, FCAI: Three-year doctoral programme funding in artificial intelligence, specialising in engineering, factory automation, and robotics.

AI-PRISM project, European Union: Contributed as a researcher to the Open Alliance Pilot, a multimodal interaction research pod for industrial human–robot collaboration.

AHEAD grant, World Bank: Contributed as a research assistant to the CSAAT project at SLIIT, focusing on culturally sensitive autism assessment using the NAO V6 robot.

Finland Government Scholarship, 2022: Merit-based scholarship supporting international students in Finland.

9. RESEARCH OUTPUT

Total number of peer-reviewed journal publications: 2.

Peer-reviewed journal articles:

- Ranasinghe, N., Mohammed, W. M., Stefanidis, K., & Martinez Lastra, J. L. (2025). *Large Language Models in Human–Robot Collaboration With Cognitive Validation Against Context-Induced Hallucinations*. IEEE Access, 13, 77418–77430. DOI: 10.1109/ACCESS.2025.3565918.
- Pham, H., Ranasinghe, N., Le, D., Atman, M. W. S., & Gusrialdi, A. (2024). *Safety-Guaranteed Distributed Formation Control of Multi-Robot Systems Over Graphs With Rigid and Elastic Edges*. IEEE Robotics and Automation Letters, 9(11), 10240–10247. DOI: 10.1109/LRA.2024.3471460.

Selected research and development outputs:

- Multi-modal interaction system for next-generation human–robot collaboration, Tampere University, 2024.
- Distributed formation control of multi-robot formations using flexible edges, Tampere University, 2023.
- Contextual Generative Adversarial Networks based face sketch-to-image synthesis, Curtin University, 2020.
- Automated ventilation system for greenhouses, Curtin University, 2020.
- Crazy Machine Project, Curtin University, 2020.
- BRANDIX Net Zero Carbon Status Project, SLIIT, 2019.

10. RESEARCH SUPERVISION AND LEADERSHIP EXPERIENCE

Thesis supervision and mentoring: Mentor / co-supervisor of Master's thesis:

Haikonen, Aaro (2026). *Reliable and Configurable Human–Robot Collaboration Framework Based on Hand Gesture Recognition Using Emoji-Guided Vision-Language Models*. Master's Programme in Automation Engineering, Tampere University.

Available at: URN:NBN:fi:tuni-202602162523.

Role included guidance on system architecture, multimodal interaction design, and integration of vision–language models in human–robot collaboration. Informal mentoring / co-supervision; not officially appointed as primary supervisor.

11. TEACHING MERITS

Academic Instructor	February 2022 – July 2022
Sri Lanka Institute of Information Technology	Malabe, Sri Lanka
– Conducted tutorials and lectures for courses in the Department of Electrical and Electronic Engineering. – Supported teaching activities, student learning, and practical engineering education.	
Robotics Workshops and Competitions	2022
Sri Lanka Institute of Information Technology	Sri Lanka
– Contributed to organising all-island robotics competitions and workshops.	

12. AWARDS AND HONOURS

Finland Government Scholarship	August 2022
Merit-based scholarship supporting international students in Finland	
Letter of Commendation	Fall 2020
Recognition for academic excellence in the final undergraduate year at Curtin University	
Curtin Extra Recognition	Fall 2021
Recognition for exemplary leadership skills at Curtin University	

13. OTHER KEY ACADEMIC MERITS

Research profile: Human–robot collaboration, collaborative robotics, large language models, industrial automation, multi-robot systems, and multi-agent systems.

Research communities and affiliations: FAST-Laboratory, Faculty of Engineering and Natural Sciences, Tampere University.

Peer review activities: Reviewer for IEEE International Conference on Robotics and Automation (ICRA) and IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS).

Conference activity: Presented research at IEEE International Conference on Robotics and Automation (ICRA 2025, Atlanta, USA) in connection with an IEEE Robotics and Automation Letters publication. Attended European Robotics Forum (ERF 2026, Stavanger, Norway). Participating in IEEE International Conference on Robotics and Automation (ICRA 2026, Vienna, Austria) as a conference participant.

Research profiles: ORCID, IEEE Xplore author profile, Tampere University Research Portal, DBLP, and GitHub.

14. SCIENTIFIC AND SOCIETAL IMPACT

Human–robot collaboration: Development of cognitively grounded and reliable approaches to the use of large language models in collaborative robotics.

Industrial systems: Contributions to multimodal interaction and conversational cobots for next-generation manufacturing systems.

Multi-robot systems: Contributions to safety-guaranteed distributed formation control in multi-robot coordination.

Societal robotics: Contribution to culturally sensitive robotic tools for autism assessment in early childhood in the Sri Lankan context.

Research dissemination: Presentation of research at international venues, including ICRA and ERF, and development of research communication platforms, including the FAST-Laboratory website.

Supervision impact: Contribution to Master’s level supervision in multimodal human–robot collaboration using vision–language models, supporting the development of adaptive Industry 5.0 systems.

15. OTHER MERITS

Social Media Responsible / INTOzine Editor / IT Officer	August 2023 – June 2026
International Teekkari Guild of Tampere University	Tampere, Finland

Exchange Tutor	January 2023 – June 2025
TREY	Tampere, Finland

Editor	December 2017 – June 2020
Rotaract Club of Achievers Lanka Business School	Colombo, Sri Lanka

Additional professional skills: Research communication, student tutoring, event organisation, social media communication, teamwork in international research environments, and interdisciplinary collaboration.

PRIVACY STATEMENT

I hereby authorize the use of my personal data in accordance with the General Data Protection Regulation, GDPR 679/16.

Signature

Date: 25.05.2026